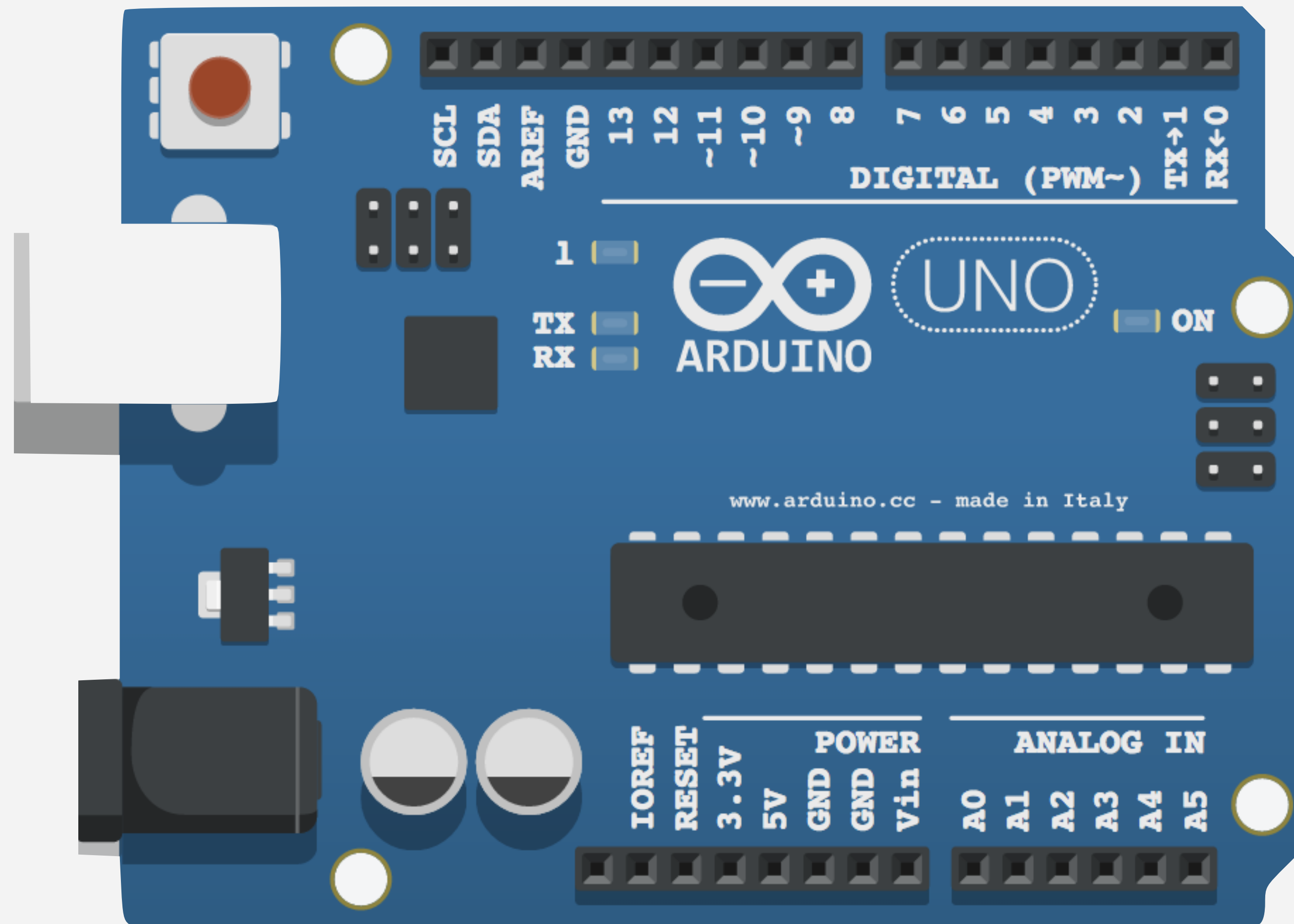
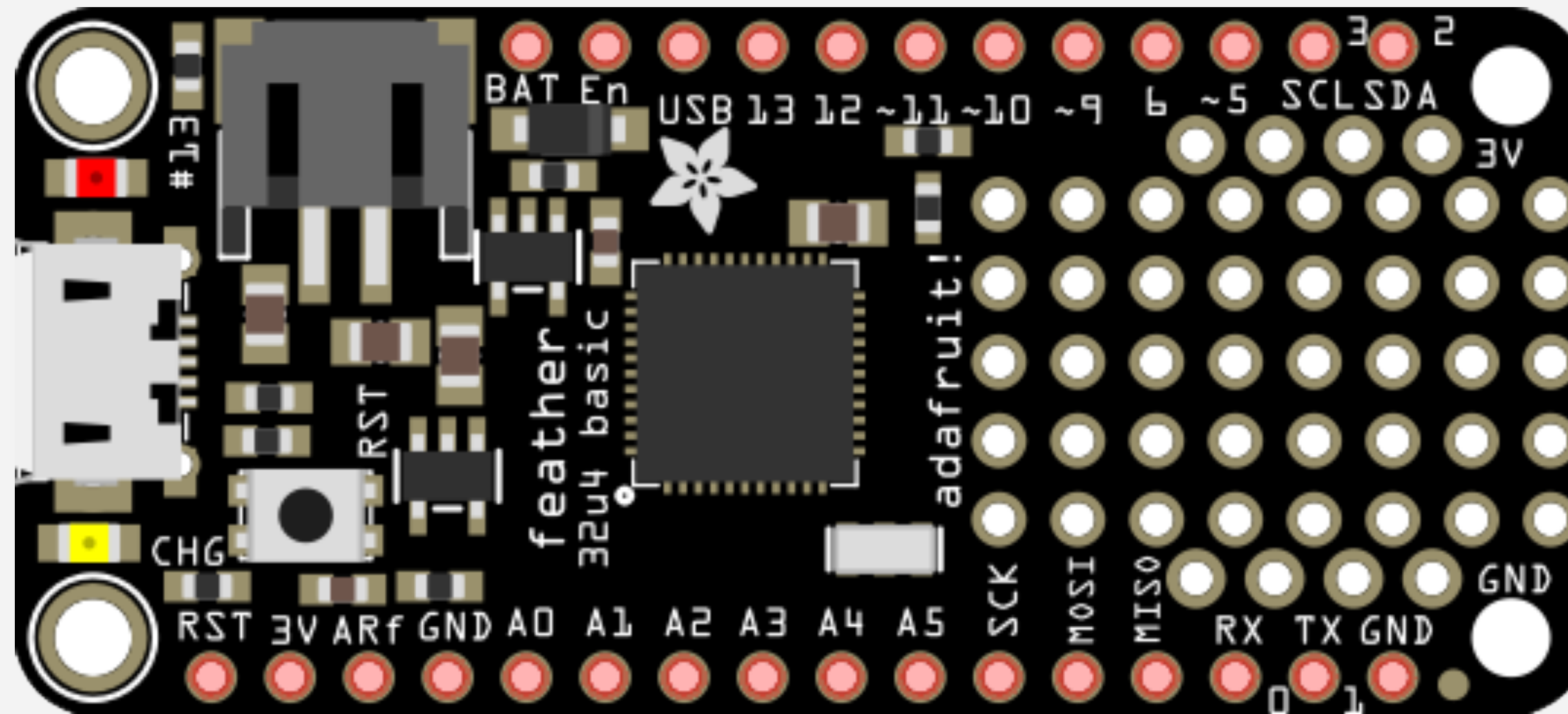


ARDUINO



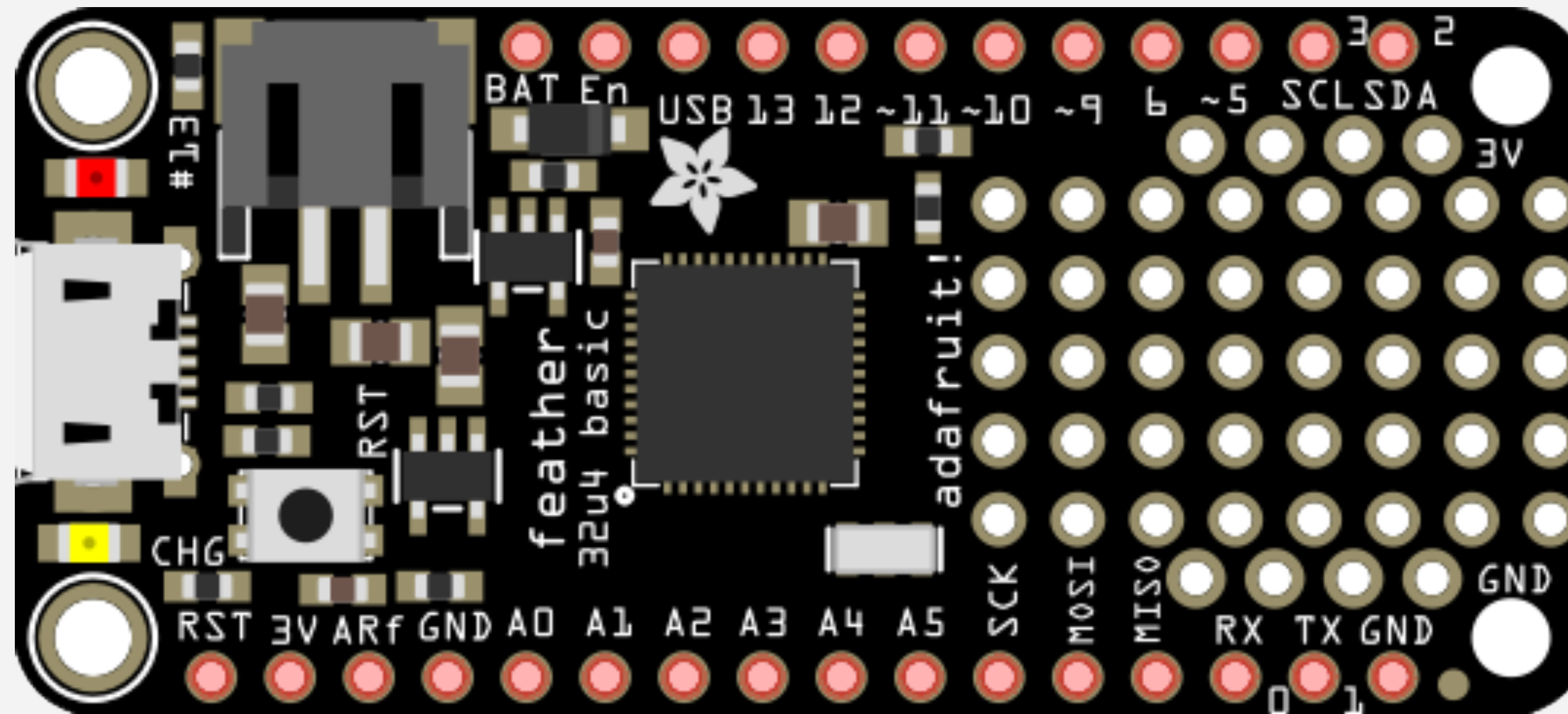
Arduino Uno

~~ARDUINO~~
FEATHER 32U4



Feather 32U4

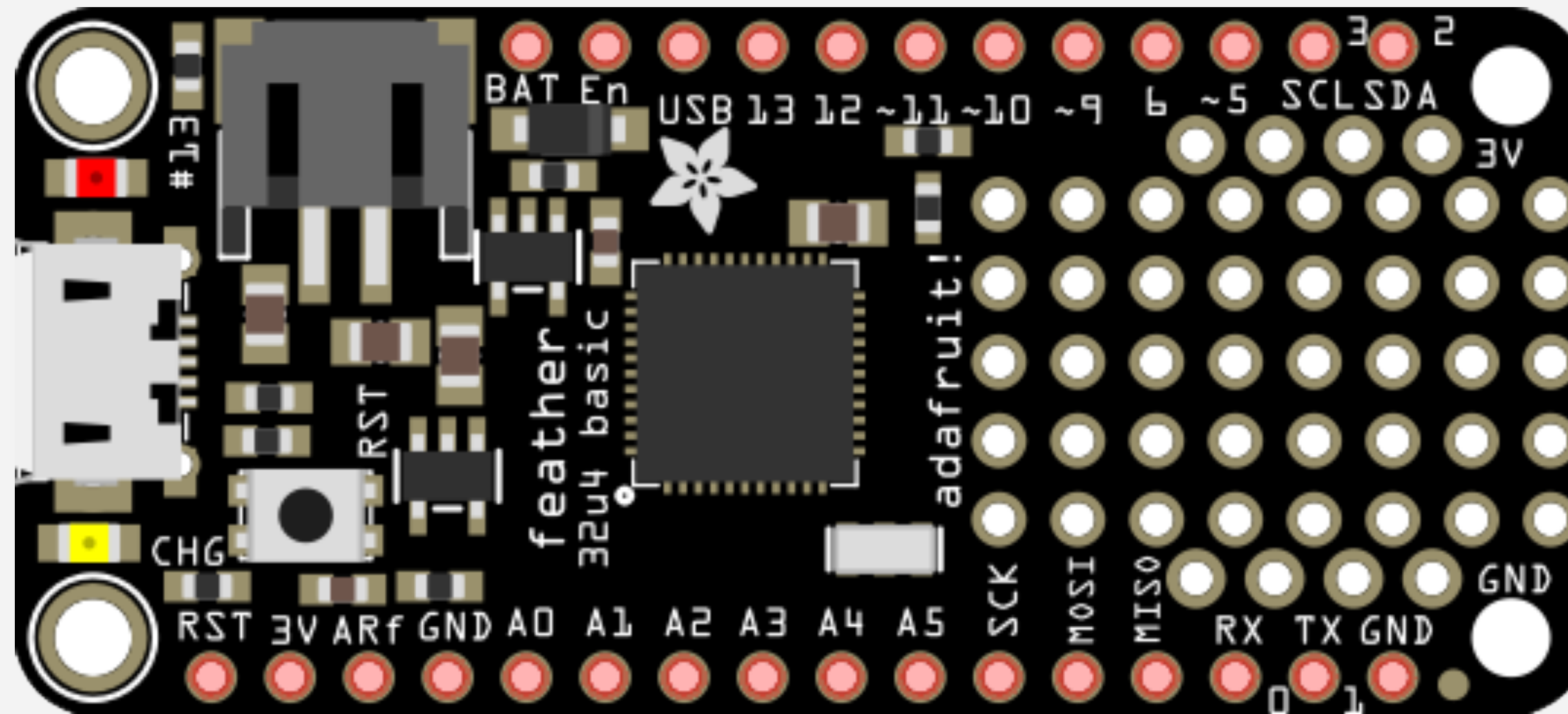
Pinout



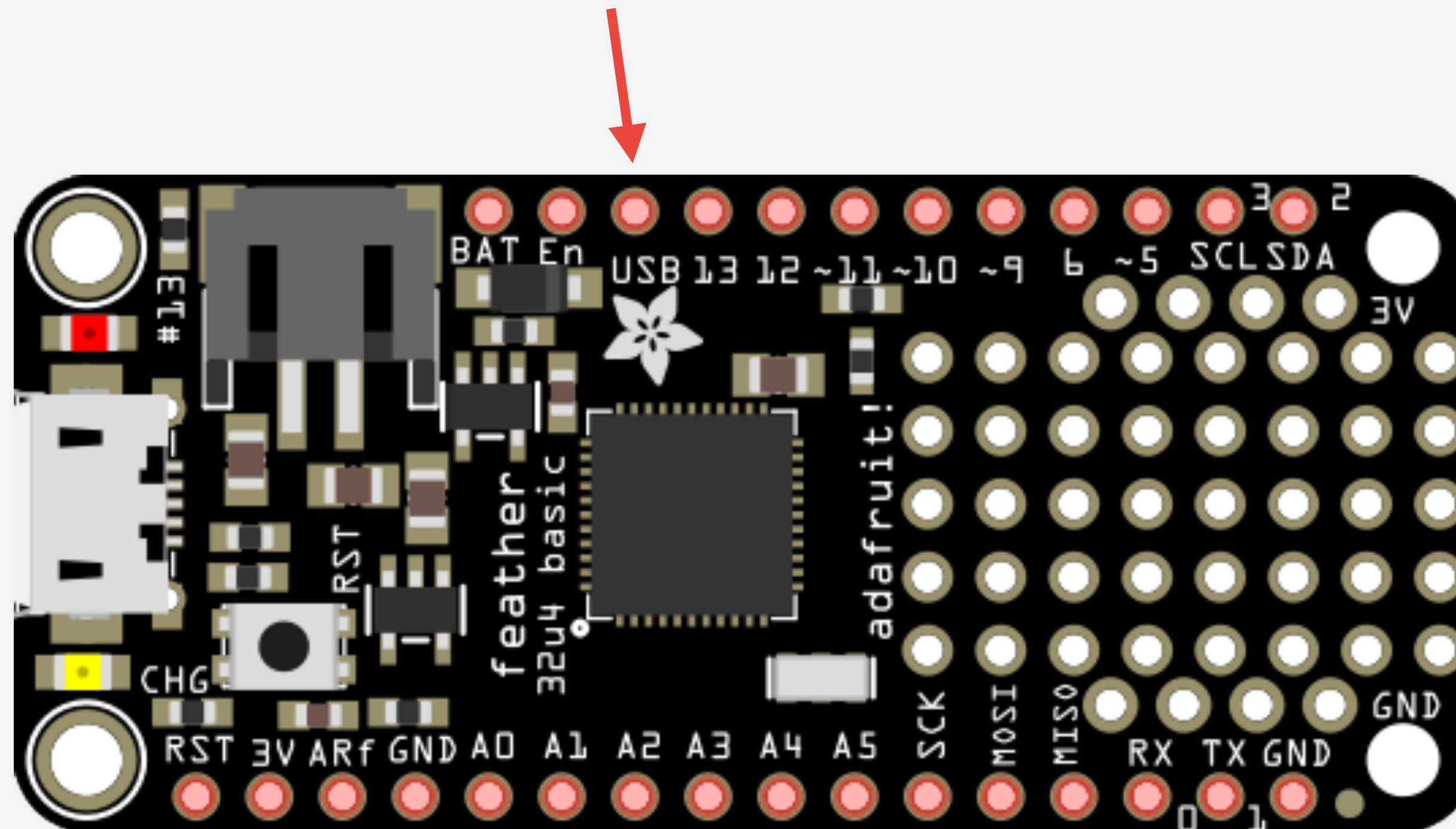
Feather 32U4



USB Port

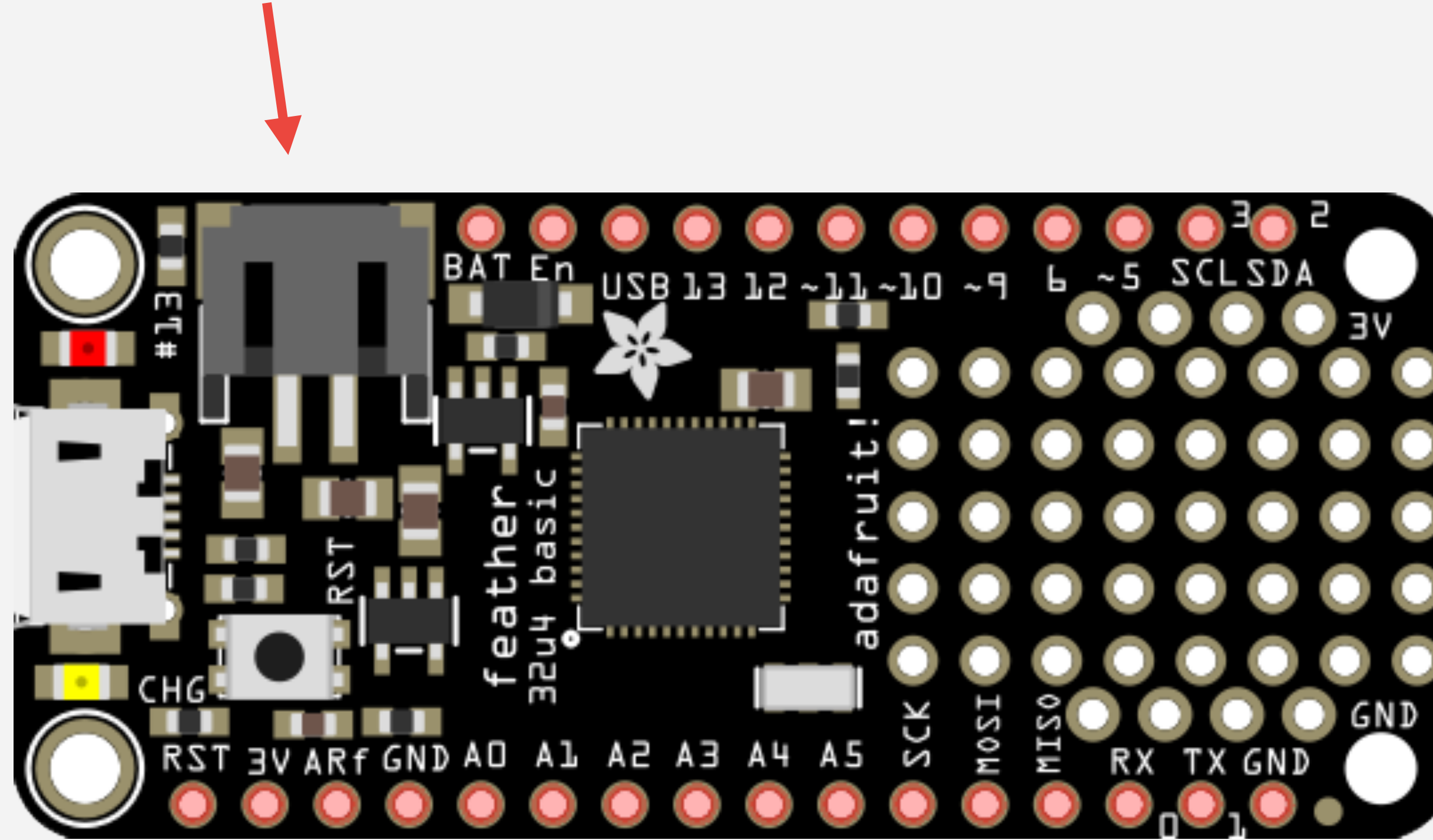


3.3V Power Output

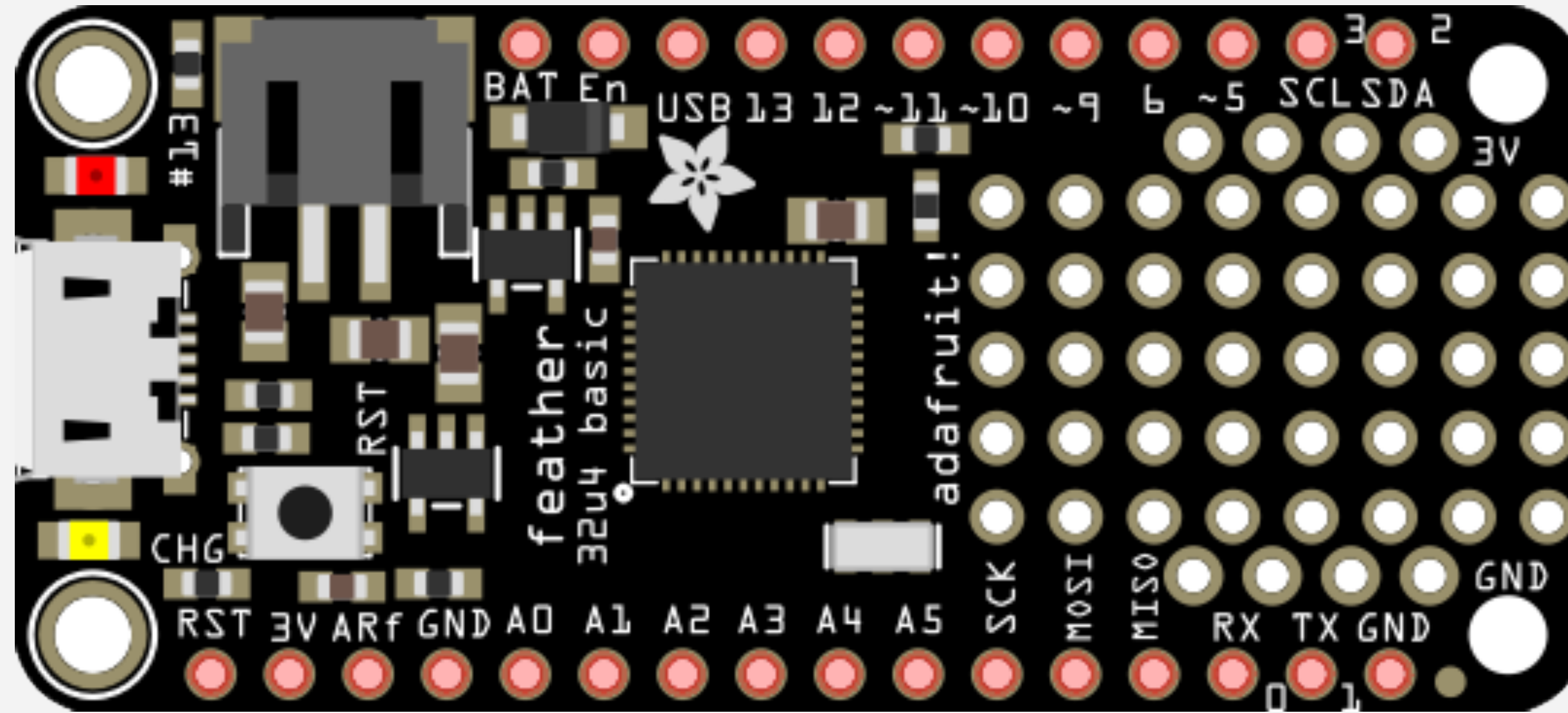


5V Power Output*

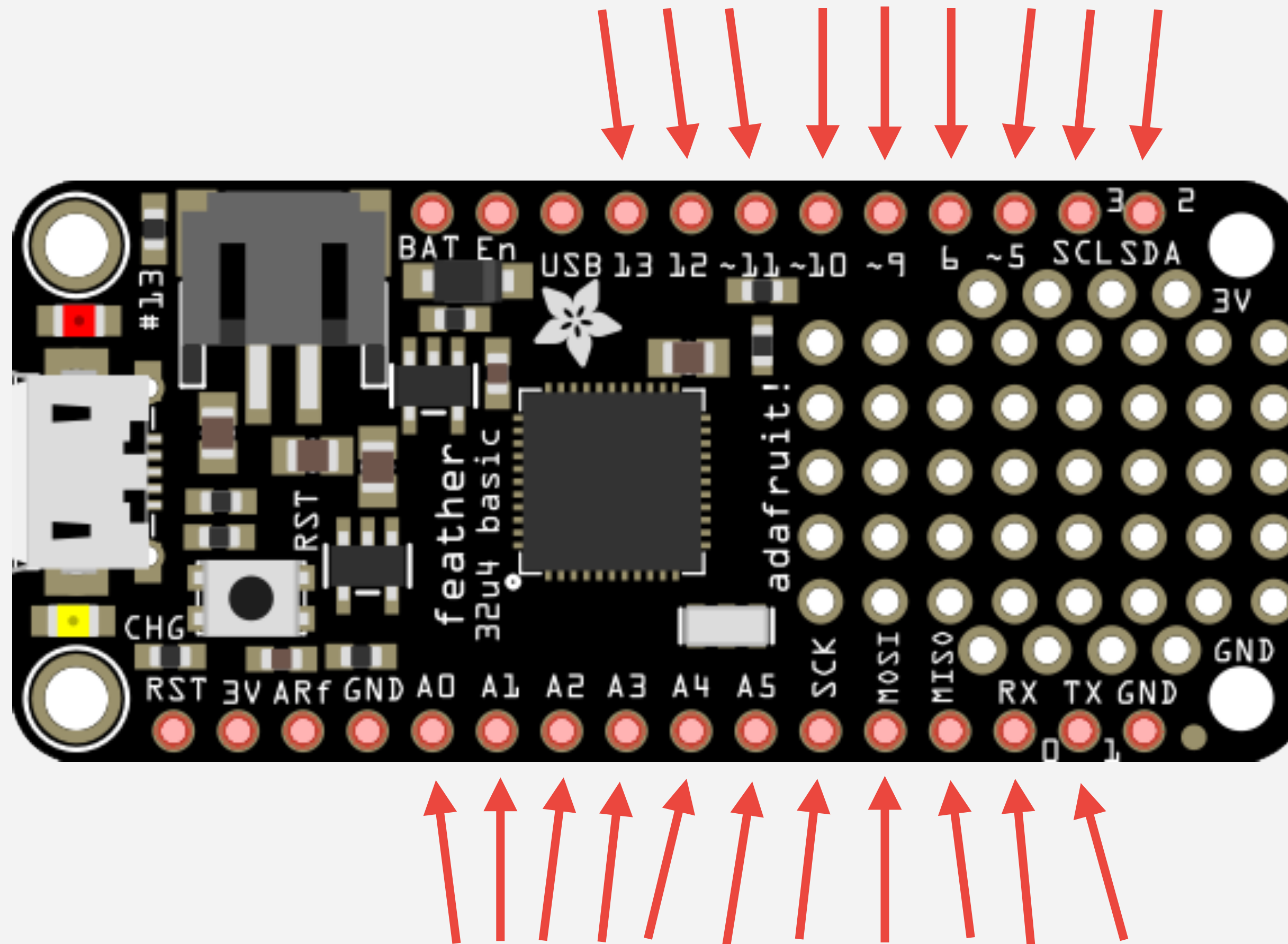
*When plugged in to USB



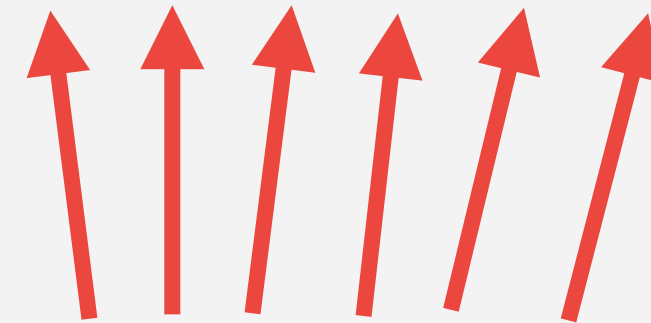
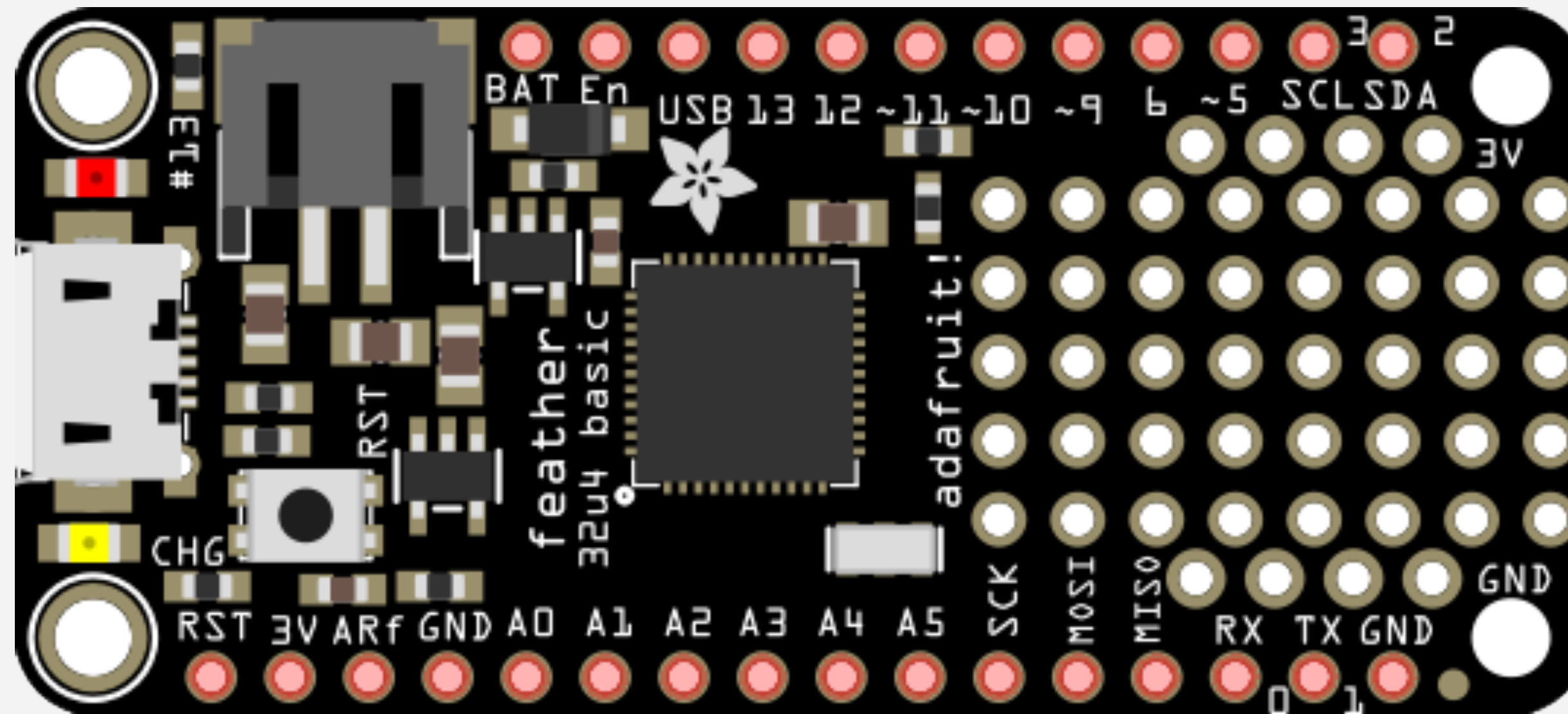
Battery Connector



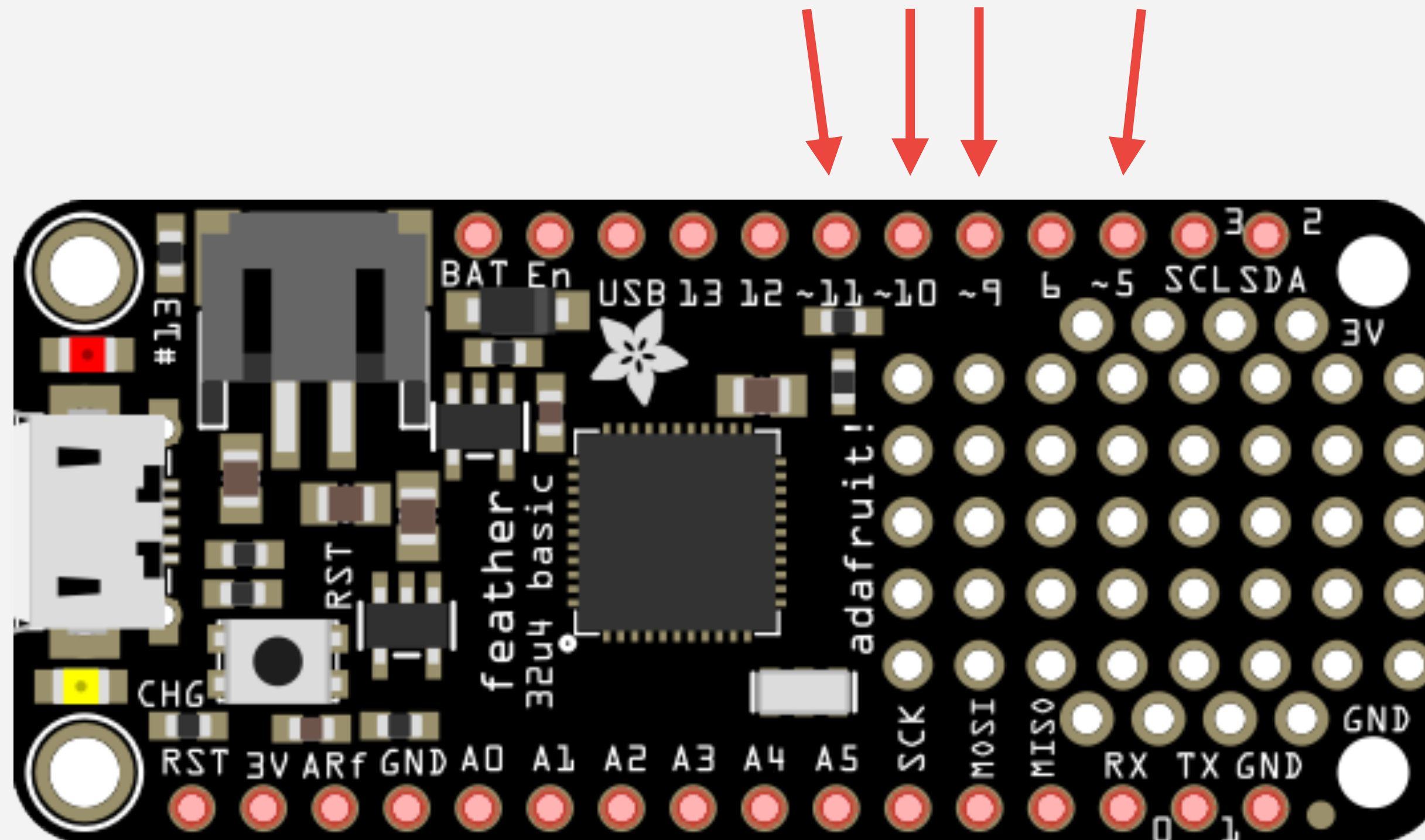
Ground



Digital Input and Output

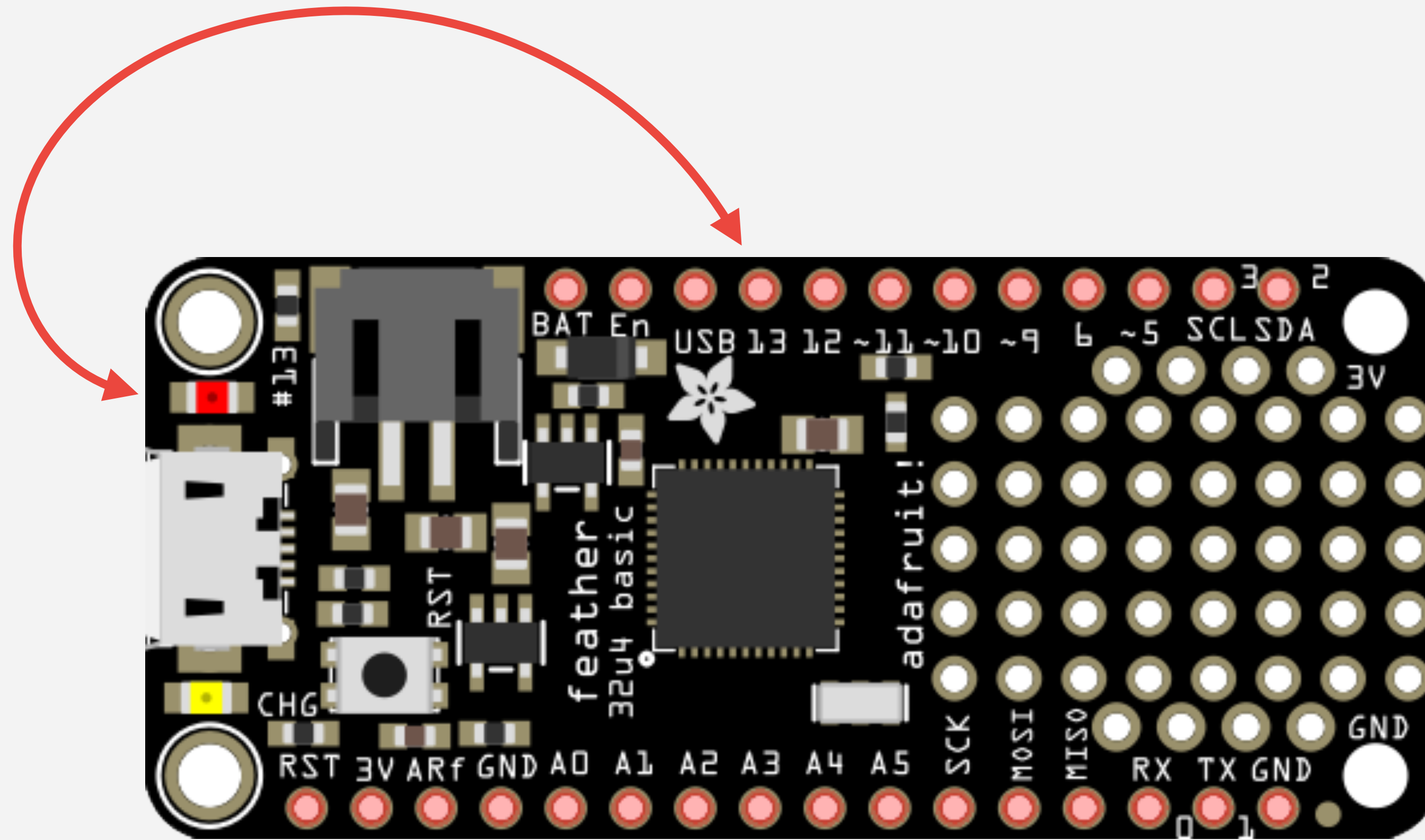


Analog Input



Analog Output*

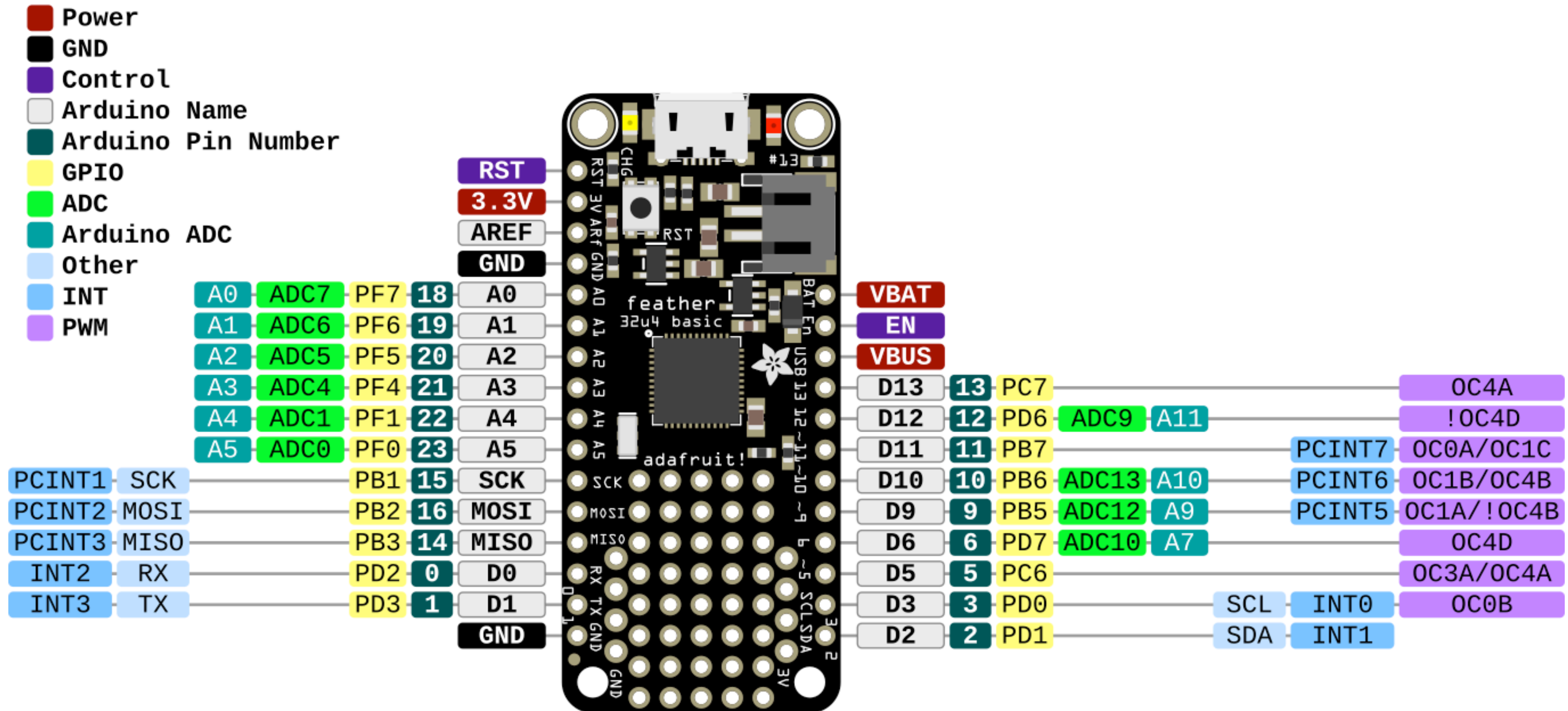
*Only on the pins with the ~ next to them



Pin 13 is connected to an LED

Adafruit Feather 32u4 Basic Proto

<https://www.adafruit.com/products/2771>



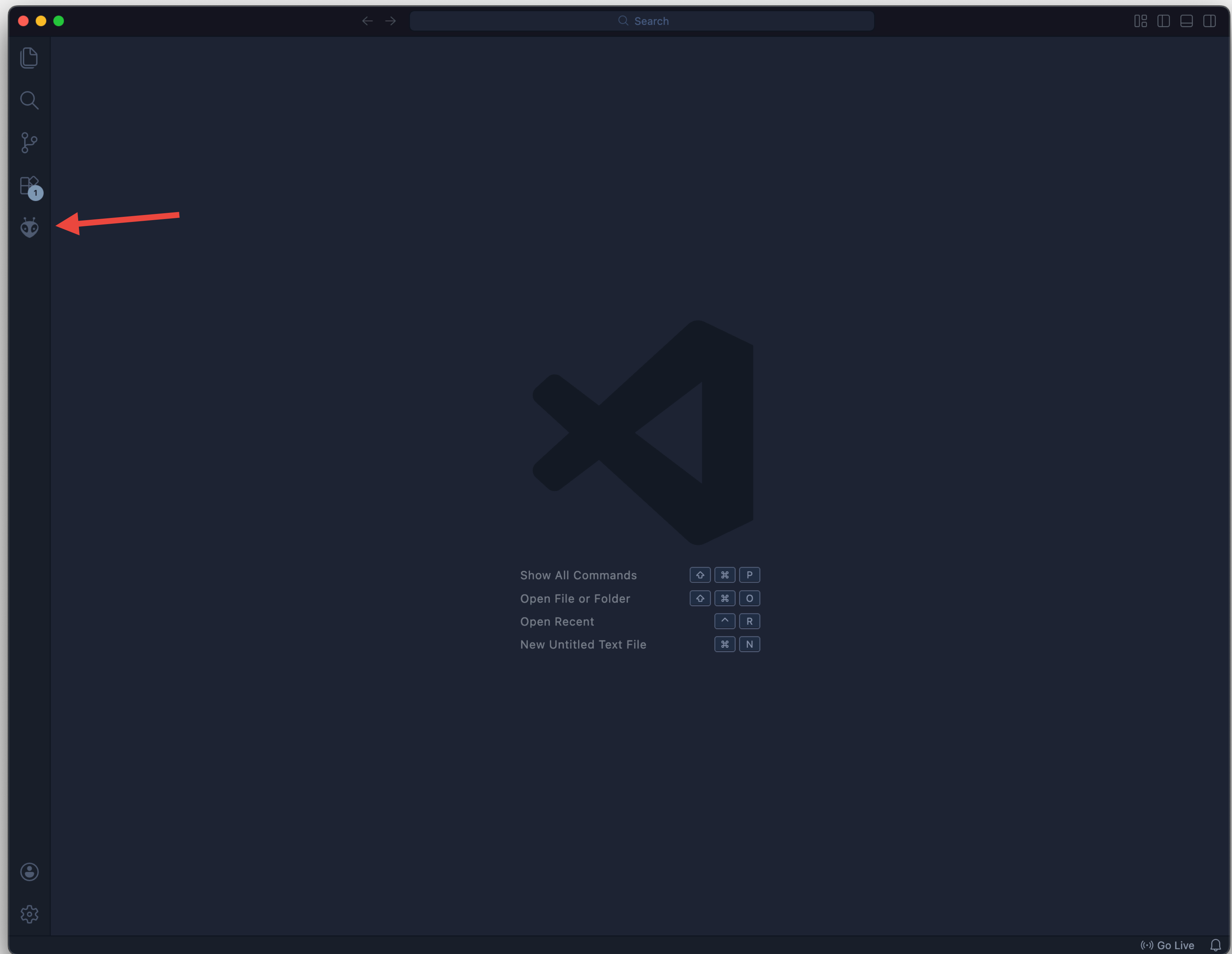
Tip!

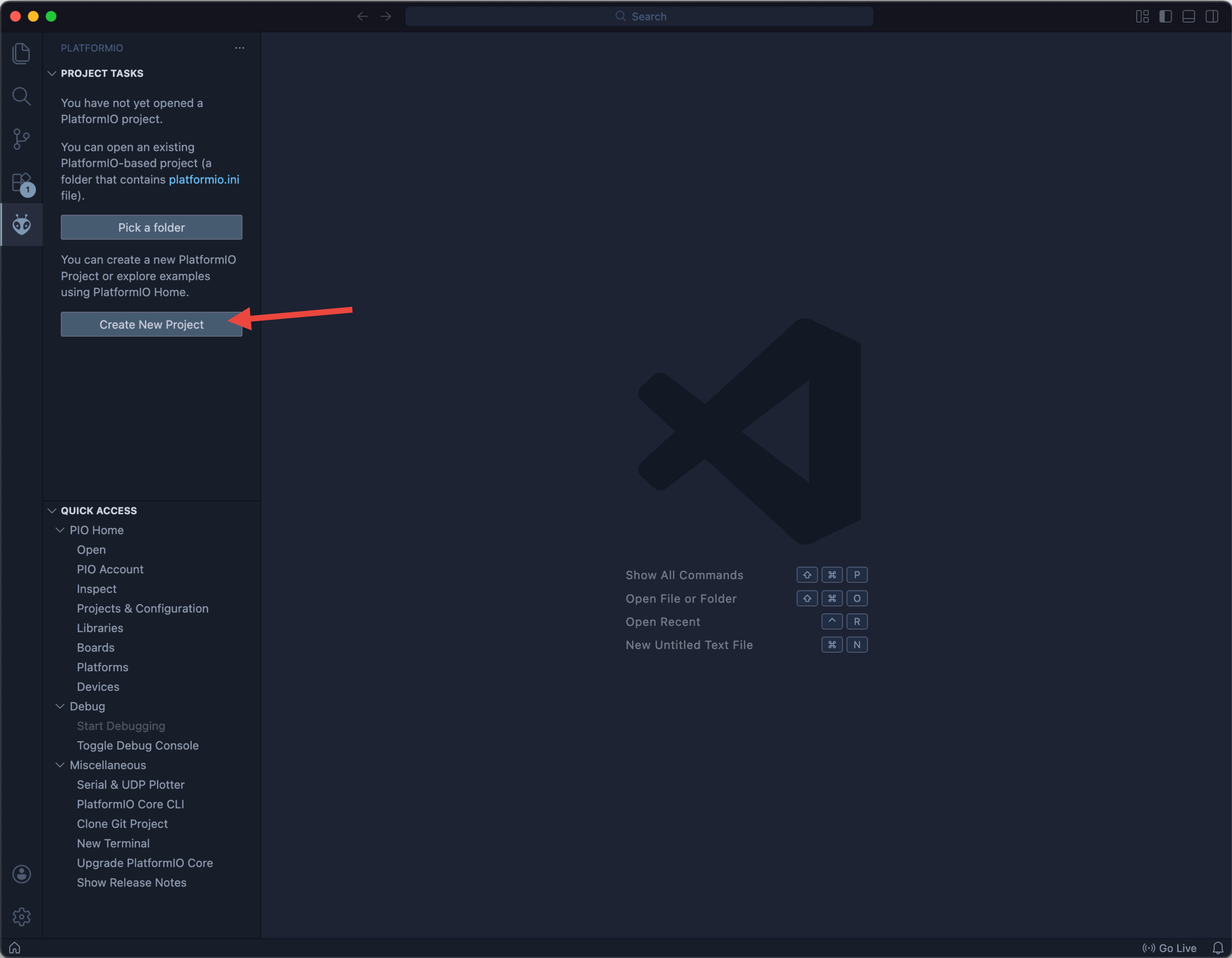
ALWAYS unplug the board
from USB before
connecting or disconnecting
things from the pins.

Setup

Install the ~~Arduino IDE~~
PlatformIO with
Feather Support

Create a new PlatformIO Project





PLATFORMIO

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Home

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Inspect

Libraries

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#LearnEmbedded "Understanding UART"

Explore the simplicity of #UART, a key communication protocol in embedded systems and computers. With just two wires, it enables

PlatformIO Labs · 6d

PlatformIO's 2025 Year in Review

★ 6M+ installs of the IDE for VSCode

🔧 20+ releases with long-term support

📦 Petabytes of traffic on our Trusted Package

PlatformIO Labs · 1w

#LearnEmbedded "Exploring Startup Implementations: Newlib (ARM)" by Phillip Johnston

Explore the program startup process and its

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Projects & Configuration

Libraries

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Debug

Start Debugging

Toggle Debug Console

Miscellaneous

Serial & UDP Plotter

PlatformIO Core CLI

Clone Git Project

New Terminal

Upgrade PlatformIO Core

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PIO Home

Home

Projects

Inspect

Libraries

Boards

Platforms

Devices

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Project Wizard

This wizard allows you to **create new** PlatformIO project or **update existing**. In the last case, you need to uncheck "Use default location" and specify path to existing project.

Name:Project name

Board:Select a board (1709 available)

Framework:

Location:☒ Use default location

Cancel

Finish

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Petabytes of traffic on our Trusted Package

Explore the program startup process and its

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Inspect

Projects & Configuration

Libraries

Boards

Platforms

Devices

Debug

Start Debugging

Toggle Debug Console

Miscellaneous

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PlatformIO Core CLI

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New Terminal

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Home

Projects

Inspect

Libraries

Boards

Platforms

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TX

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Name:

Board:

Framework:

Location: ☒ Use default location

Cancel Finish

Go Live

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Libraries

Boards

Platforms

Devices

Debug

Start Debugging

Toggle Debug Console

Miscellaneous

Serial & UDP Plotter

PlatformIO Core CLI

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PIO Home

Home

Projects

Inspect

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Framework:

Location: ☒ Use default location ?

Cancel

Finish

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Explore the program startup process and its

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Search project...

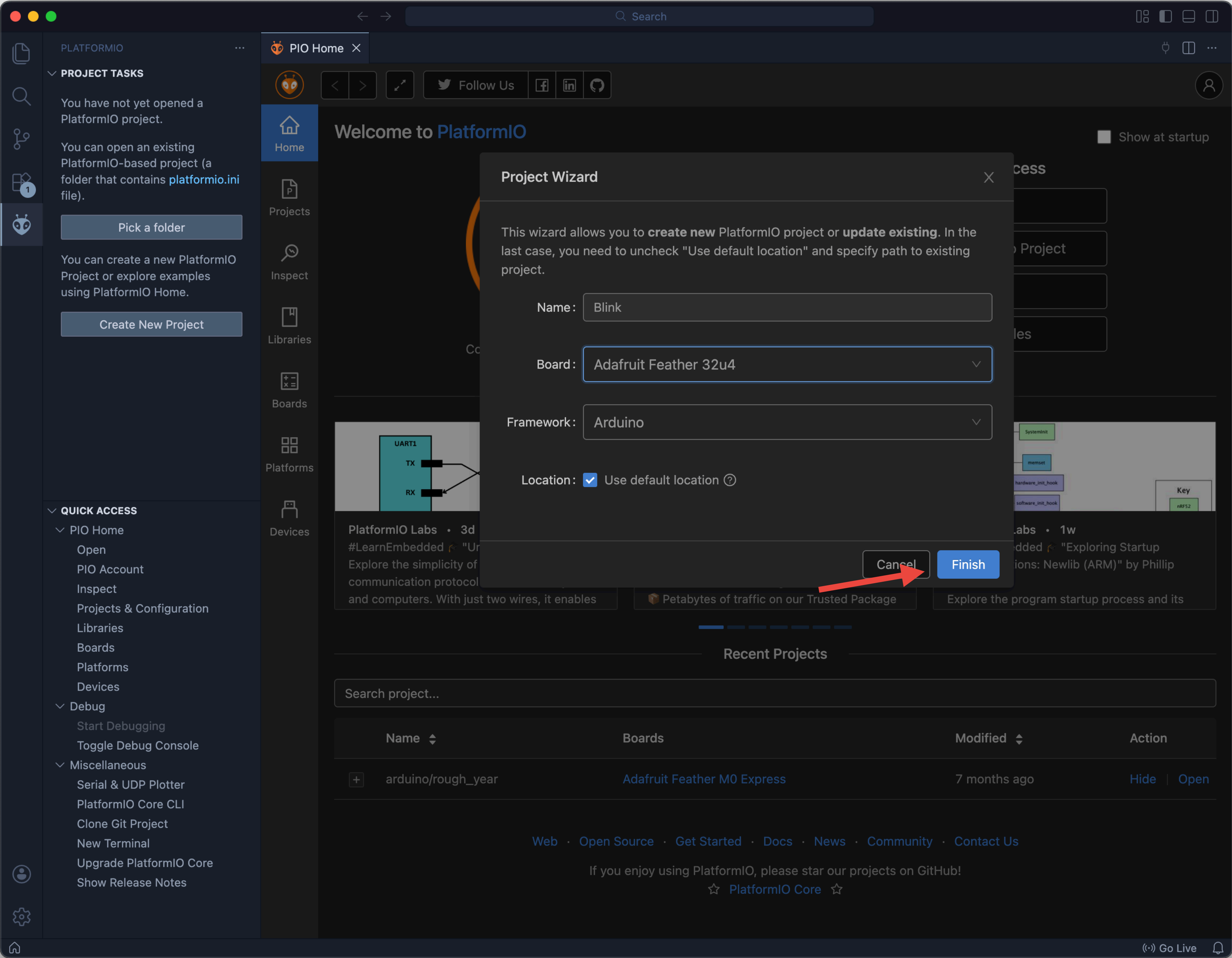
Name	Boards	Modified	Action
+ arduino/rough_year	Adafruit Feather M0 Express	7 months ago	Hide Open

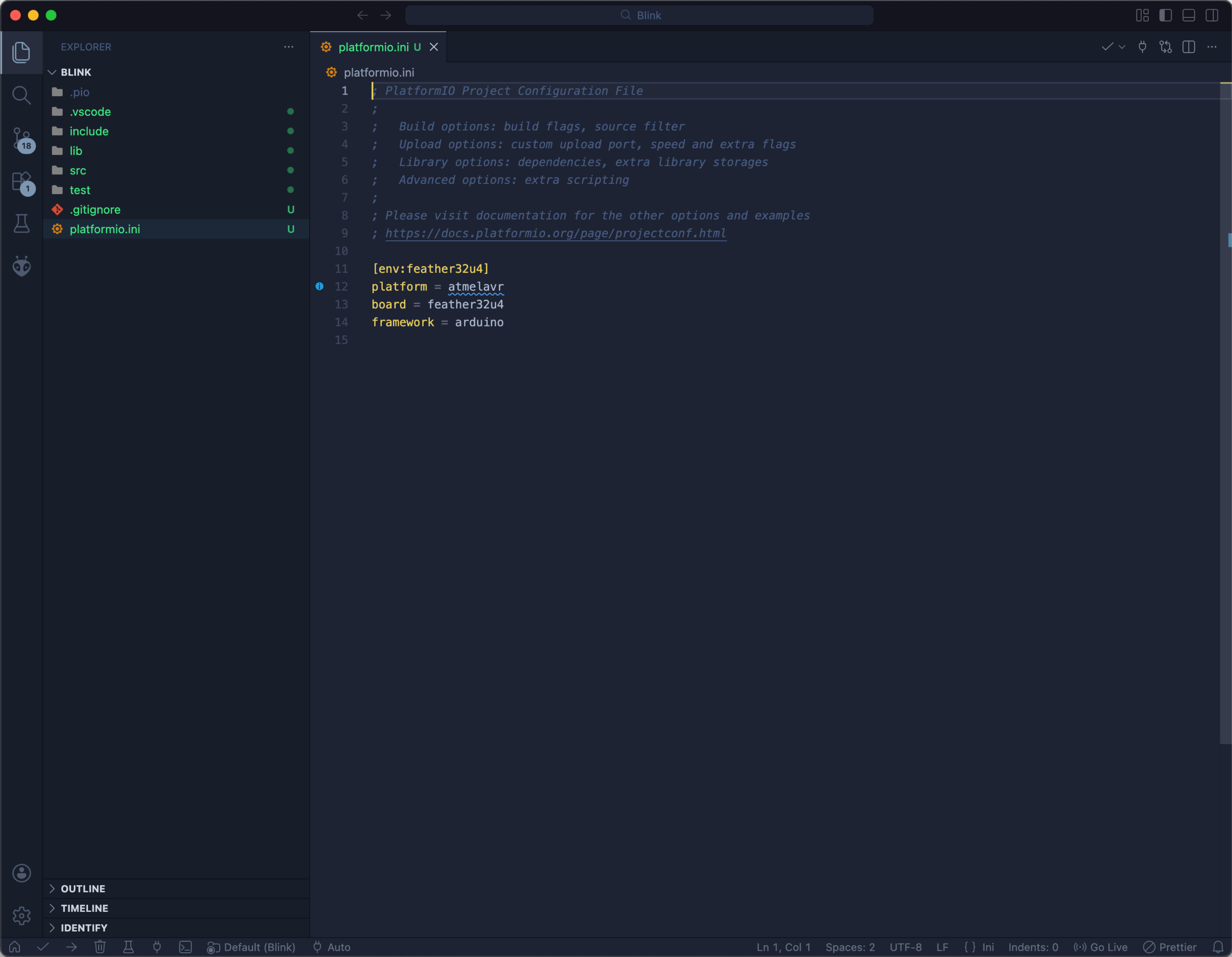
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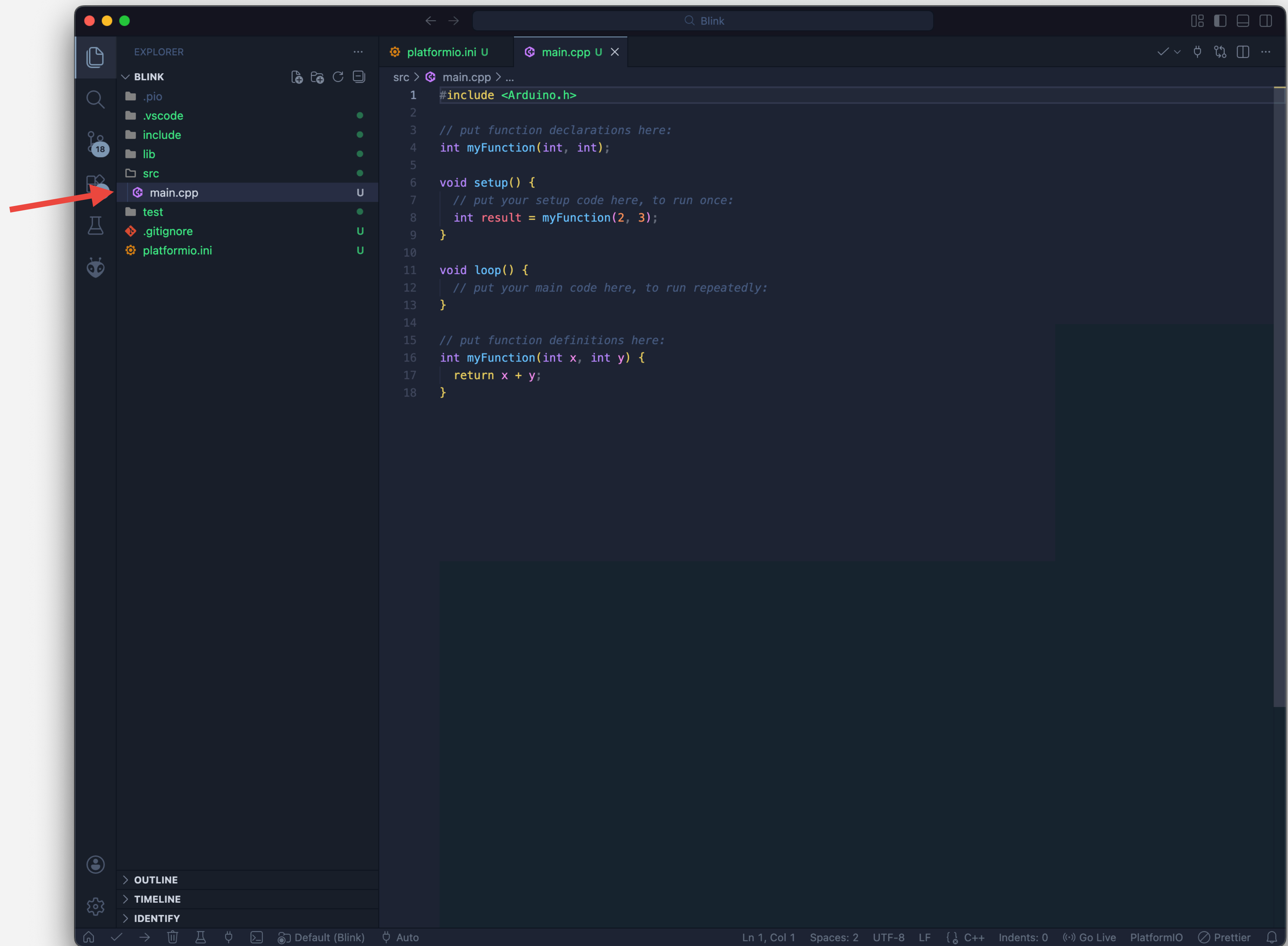
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☆ PlatformIO Core ☆

Go Live

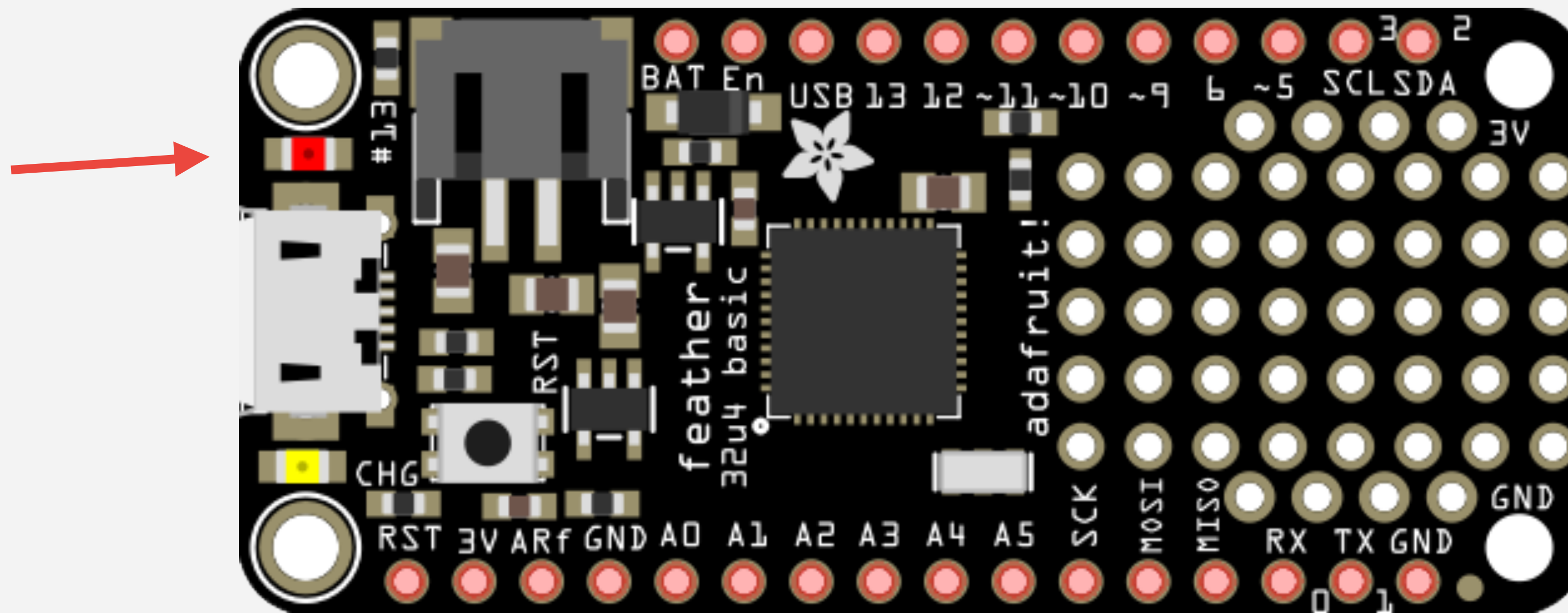




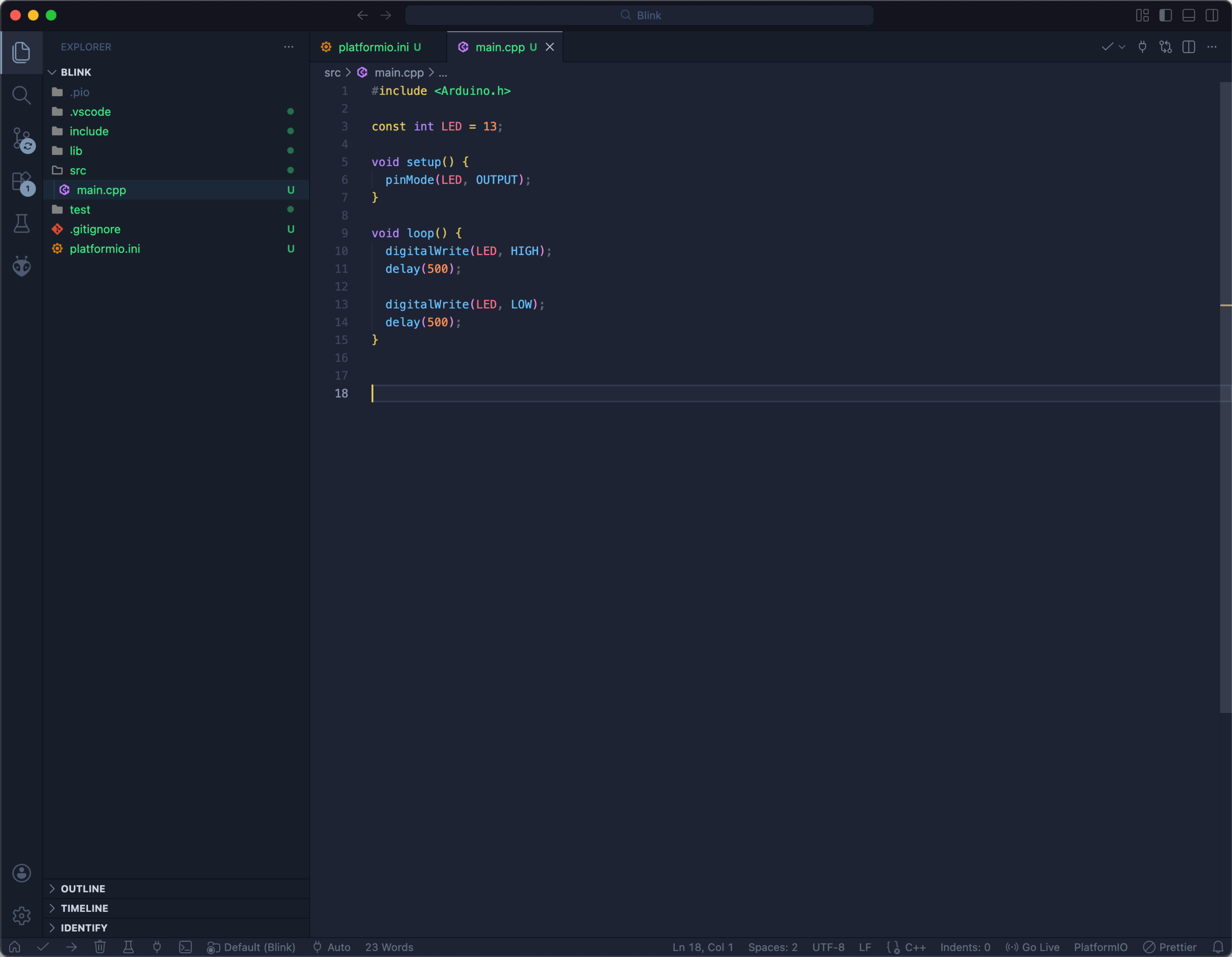


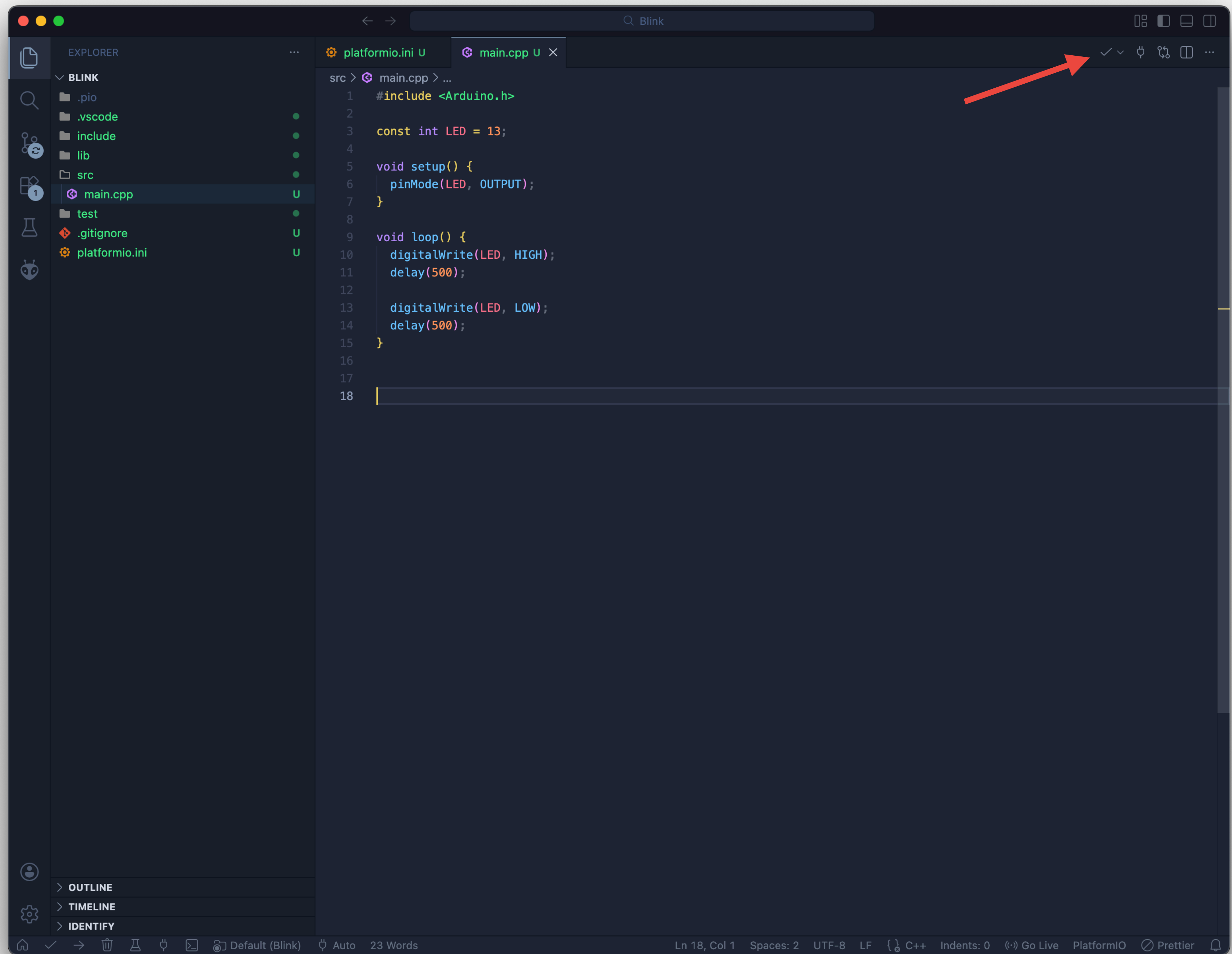
HELLO LED!

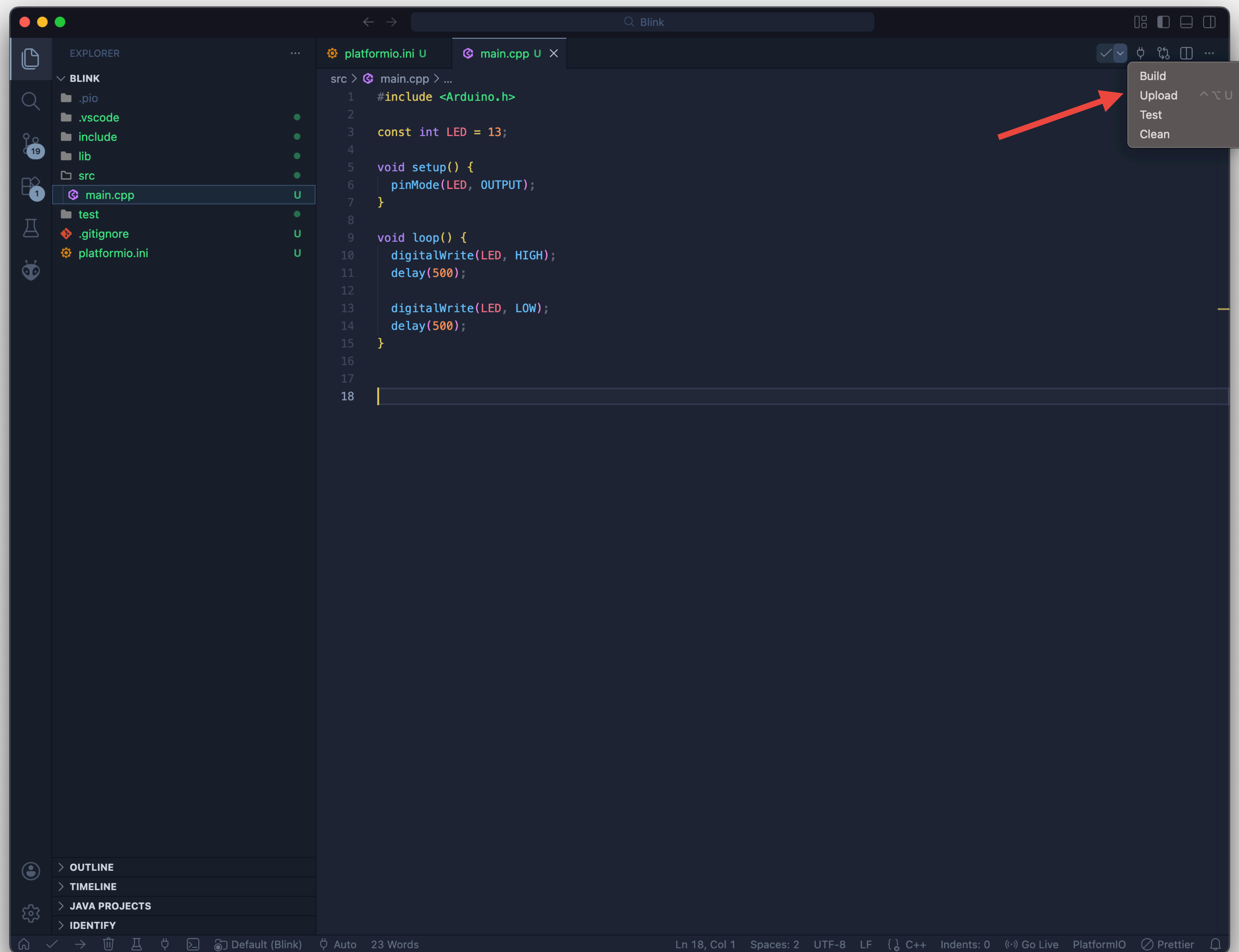
Blink the built-in LED



Built-in LED







18

1

EXPLORER

BLINK

.pio

.vscode

include

lib

src

main.cpp

test

.gitignore

platformio.ini

OUTLINE

TIMELINE

JAVA PROJECTS

IDENTIFY

PlatformIO: Upload

src > main.cpp > LED

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

```
#include <Arduino.h>

const int LED = 13;

void setup() {
  pinMode(LED, OUTPUT);
}

void loop() {
  digitalWrite(LED, HIGH);
  delay(500);

  digitalWrite(LED, LOW);
  delay(500);
}
```

TERMINAL 1

TERMINAL

* Executing task: platformio run

* The terminal process "platformio 'run'" terminated with exit code: 1.

* Terminal will be reused by tasks, press any key to close it.

* Executing task: platformio run

Processing feather32u4 (platform: atmelavr; board: feather32u4; framework: arduino)

Verbose mode can be enabled via '-v, --verbose' option

CONFIGURATION: <https://docs.platformio.org/page/boards/atmelavr/feather32u4.html>

PLATFORM: Atmel AVR (5.1.0) > Adafruit Feather 32u4

HARDWARE: ATMEGA32U4 8MHz, 2.50 KB RAM, 28KB Flash

DEBUG: Current (simavr) External (simavr)

PACKAGES:

- framework-arduino-avr @ 5.2.0

- toolchain-atmelavr @ 1.70300.191015 (7.3.0)

LDF: Library Dependency Finder -> <https://bit.ly/configure-pio-ldf>

LDF Modes: Finder ~ chain, Compatibility ~ soft

Found 5 compatible libraries

Scanning dependencies...

No dependencies

Building in release mode

Compiling .pio/build/feather32u4/src/main.cpp.o

Archiving .pio/build/feather32u4/libFrameworkArduino.a

Indexing .pio/build/feather32u4/libFrameworkArduino.a

Linking .pio/build/feather32u4/firmware.elf

Checking size .pio/build/feather32u4/firmware.elf

Advanced Memory Usage is available via "PlatformIO Home > Project Inspect"

RAM: [] 5.8% (used 149 bytes from 2560 bytes)

Flash: [] 13.5% (used 3862 bytes from 28672 bytes)

Building .pio/build/feather32u4/firmware.hex

= [SUCCESS] Took 0.73 seconds =

* Terminal will be reused by tasks, press any key to close it.

PROBLEMS

OUTPUT

1

Default (Blink)

Auto

23 Words

Ln 3, Col 19

Spaces: 2

UTF-8

LF

C++

Indents: 0

Go Live

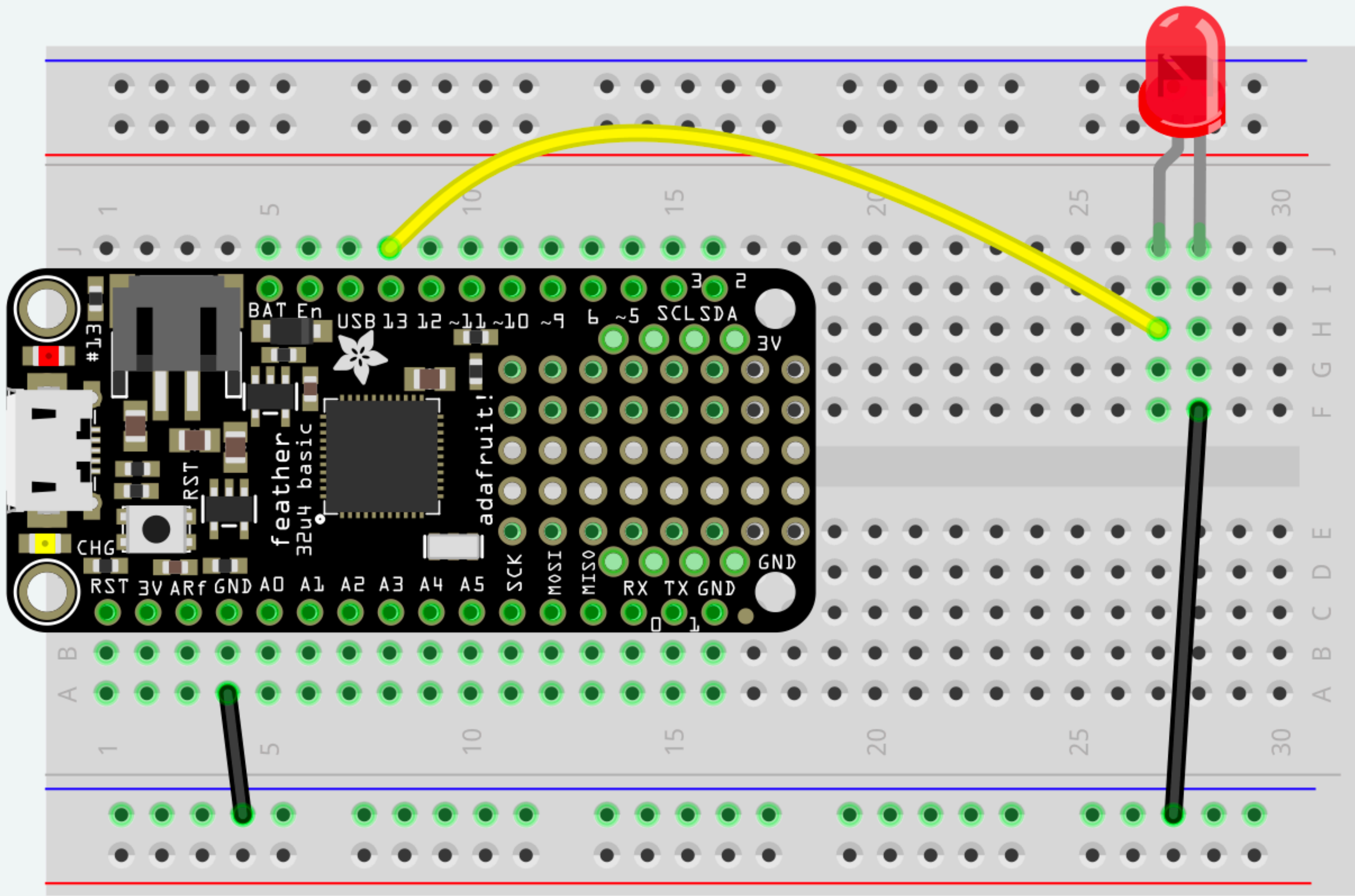
PlatformIO

Prettier

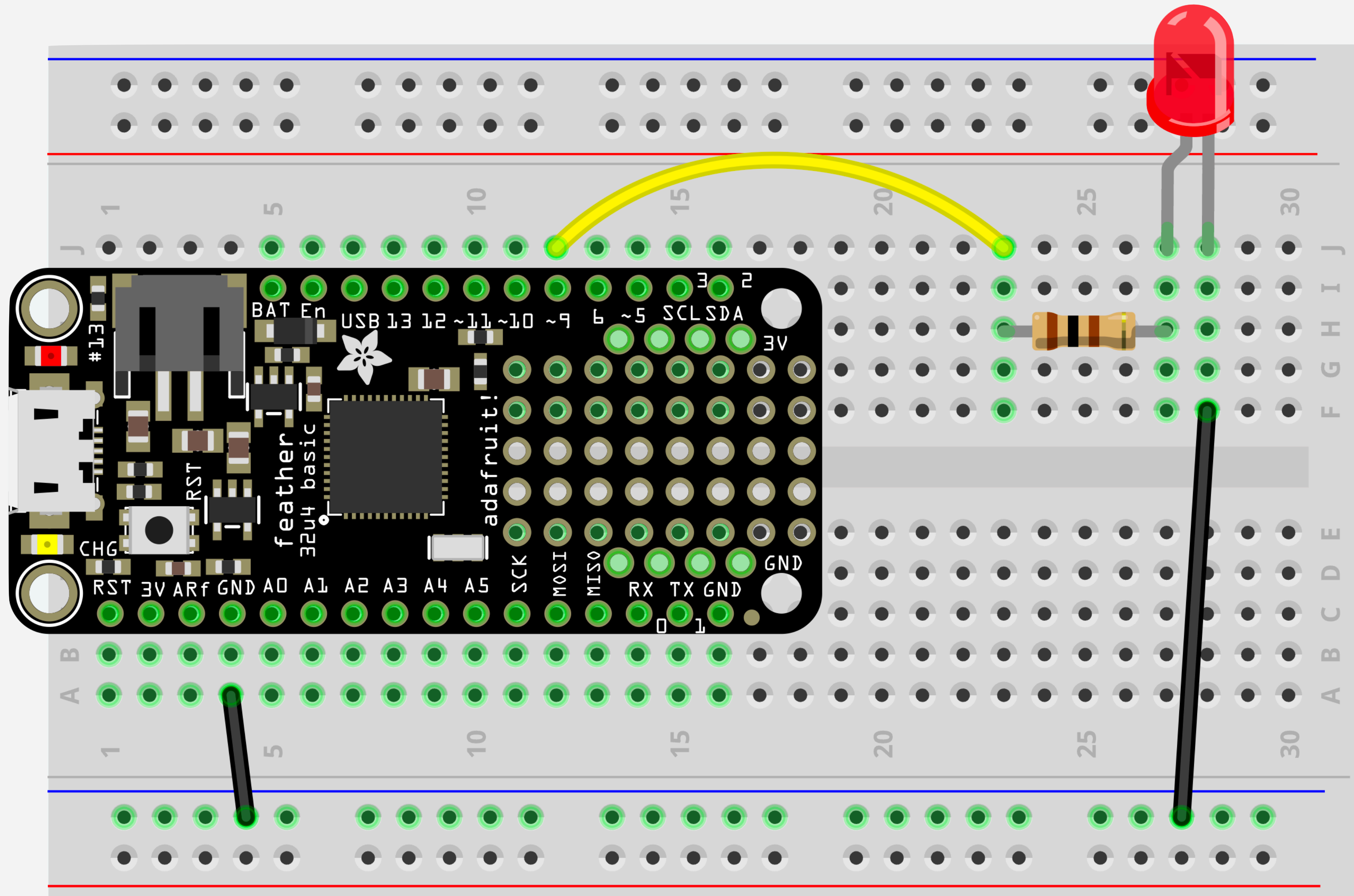
Try changing the blink frequency,
then try creating a pattern.

Remember that the `delay()`
function is what makes the code
pause.

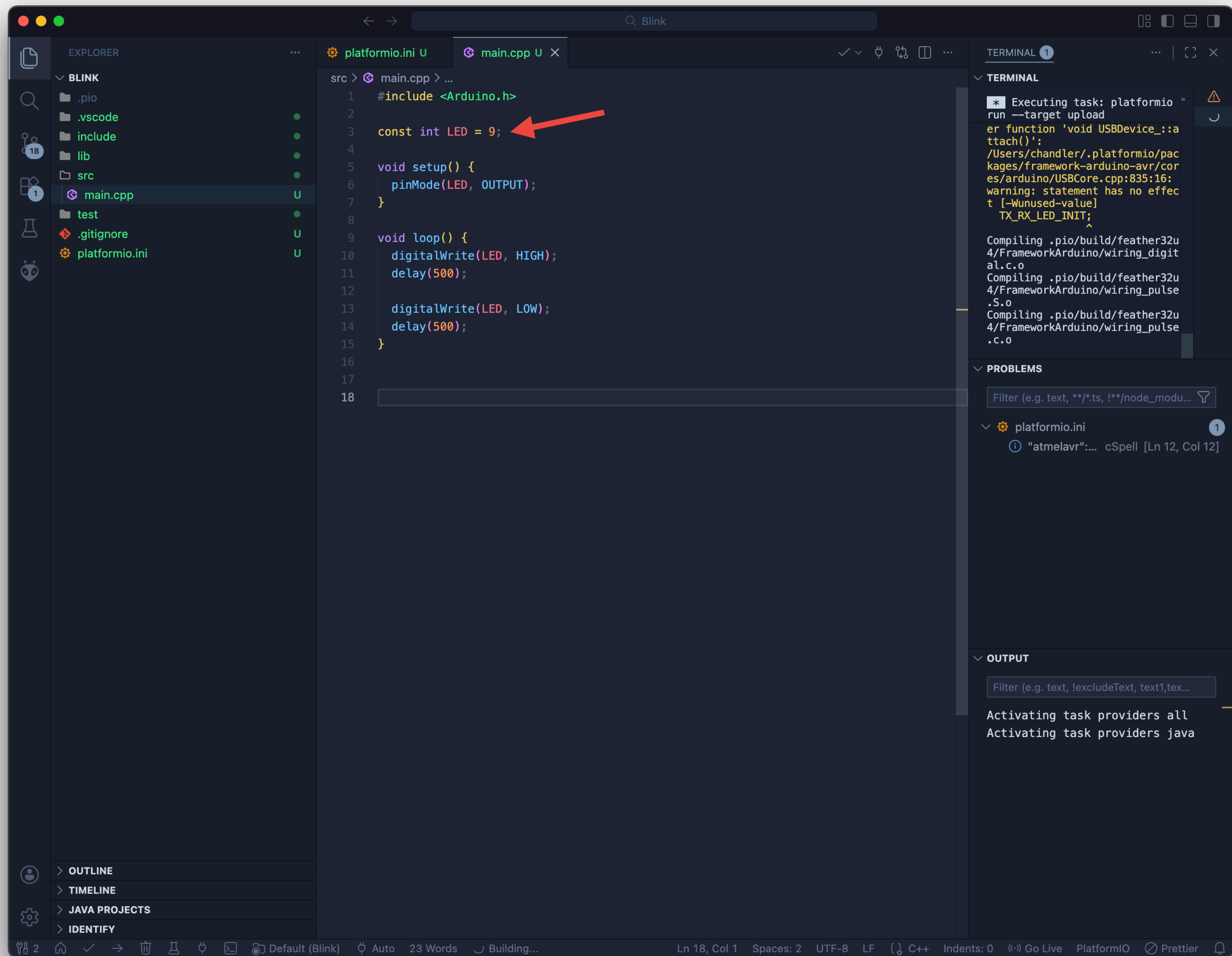
Blink an external LED on Pin 13



**Blink an external LED
on other pins**



* Resistor is 100Ω (Brown Black Brown)

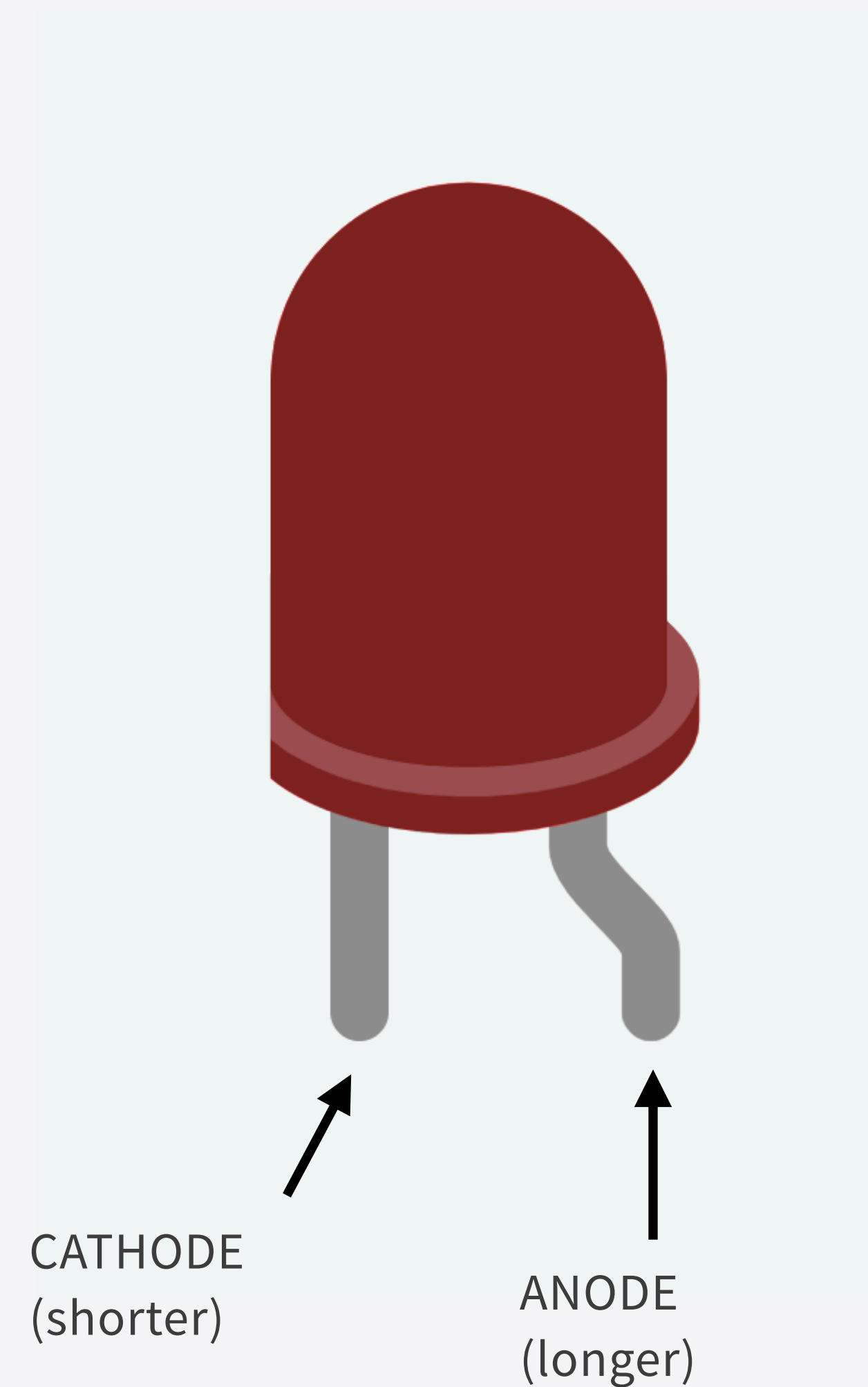


Try connecting more LEDs
to other pins.

What patterns can you create?

What limits/complications are caused
by using the `delay()` function?

An LED has two legs, a longer one called an anode and a shorter one called a cathode



Short Circuit!

This can happen by accidentally connecting the positive and negative terminals of a battery, or, more often, but some loose wires coming out and touching.

Make sure your wires are always trimmed and minimize the amount of metal sticking out.

